

Adeeb Khan

Senior Software Engineer @ Marvell Technology
Systems | Backend | AI

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[GitHub](#) [in LinkedIn](#) [Website](#)

Education

Motilal Nehru National Institute of Technology Allahabad, Prayagraj

M.Tech (Computer Science & Engineering)

Aug. 2023 - June 2025

CPI: 9.0

Bundelkhand Institute of Engineering and Technology, Jhansi

B.Tech (Computer Science & Engineering)

Aug. 2018 - June 2022

CGPA: 8.54

Experience

- **Senior Software Engineer** | Marvell Technology (Jun'25 – Present)
 - Built a Python-based **VDK orchestration framework** automating multi-stage firmware setup, configuration, and patch deployment with resumable execution; adopted across development, QA, and CI teams.
 - Engineered a **libFuzzer-based firmware fuzzing framework** with simulated hardware abstractions and NVMe communication, identifying **5+ firmware crashes across 2 modules**.
 - Developed an **LLM-assisted release-validation tool** analyzing Jira fixes and regression data to recommend high-priority test scenarios with reasoning.
 - Automated **qTest result creation and log attachment** via Python and API integration; built firmware automation across **NVMe, MCTP, and PLDM**.
- **Software Engineer Intern** | Marvell Technology (Jul'24 – Jun'25)
 - Built a Python **AWS Bedrock** tool for requirement-to-test generation, improving validation efficiency by **~25%**.
 - Identified **15+ firmware defects** through automated and scenario-driven testing of SSD controller firmware.
 - Automated **50+ regression scenarios**, saving **8+ hours/week**, and enhanced diagnostics to surface relevant failure data and logs.

Research Experience

- **Early Identification of Suicide-Susceptible Students using Computational Techniques** | Prof. Divya K | MNNIT Allahabad
 - Developed a **predictive ML classification system** using **DELM, SVM, RF, and XGBoost**, achieving **79% accuracy**; led dataset creation, feature engineering, training, and evaluation as lead author.
 - Got the paper accepted at **IEEE ICONAT 2025**. [\[IEEE Xplore\]](#)
- **Fake Image Detection Research** | Prof. Sarsij Tripathi | MNNIT Allahabad
 - Built a **stacking ensemble of 5 CNNs** with multiple meta-classifiers including **RF, SVM, GBM, and MLP**, achieving **>90% accuracy** on unseen data. [\[Pre-Print\]](#)
 - Designed a **6-channel hybrid-feature CNN** combining RGB, DCT, LBP, and gradient features, achieving **97.3% AUC / 91.4% accuracy**; accepted at **IEEE ICDSINC 2025**. [\[IEEE Xplore\]](#)

Projects

- **C++ In-Memory Cache** | Systems Programming [\[GitHub Repo\]](#)
 - Built an in-memory cache using non-blocking **TCP sockets and epoll**, with an event-driven architecture, custom protocol, and **PING, SET, GET, DEL** operations.
- **Real-Time Messenger** | Full Stack Systems [\[GitHub Repo\]](#)
 - Built a real-time messaging platform using **React, Node.js, WebSockets, and MongoDB**, implementing connection management, message routing, authentication, and file sharing.
- **AI Resume Modifier CLI** | Applied AI [\[GitHub Repo\]](#)
 - Engineered a **Python CLI** using the **Google Gemini API** to parse job descriptions and LaTeX resumes, perform requirement/relevance analysis, and generate tailored resumes.
- **AI Music Generator** | Deep Learning [\[GitHub Repo\]](#)
 - Built an **LSTM-based sequence model** in TensorFlow to learn musical patterns from MIDI note/chord sequences and generate new sequences autoregressively.

Technical Skills

- **Languages:** C++, Python, JavaScript, C, Java
- **Backend & Systems:** TCP/IP, REST APIs, WebSockets, Linux, Concurrency, Multithreading, Networking
- **Databases & Infrastructure:** MongoDB, MySQL, Git, Jenkins, Gerrit, Docker
- **AI & ML:** PyTorch, TensorFlow, Keras, Deep Learning, CNNs, Transformers, LLM Applications, AWS Bedrock
- **Protocols:** NVMe, MCTP, PLDM, I²C, I³C, PCIe VDM

Achievements

- **GATE-2022:** All India Rank **2008**.
- Solved **500+ Data Structures & Algorithms problems** across LeetCode, GeeksforGeeks, and CodeChef.
- Received **Gold and Silver Performance Awards** at Marvell Technology for impactful contributions and deliverables.
- **Poster Presentation:** Selected to present an **ML-based NVMe rate-limiting concept** among **200+ posters** at Marvell Technology.